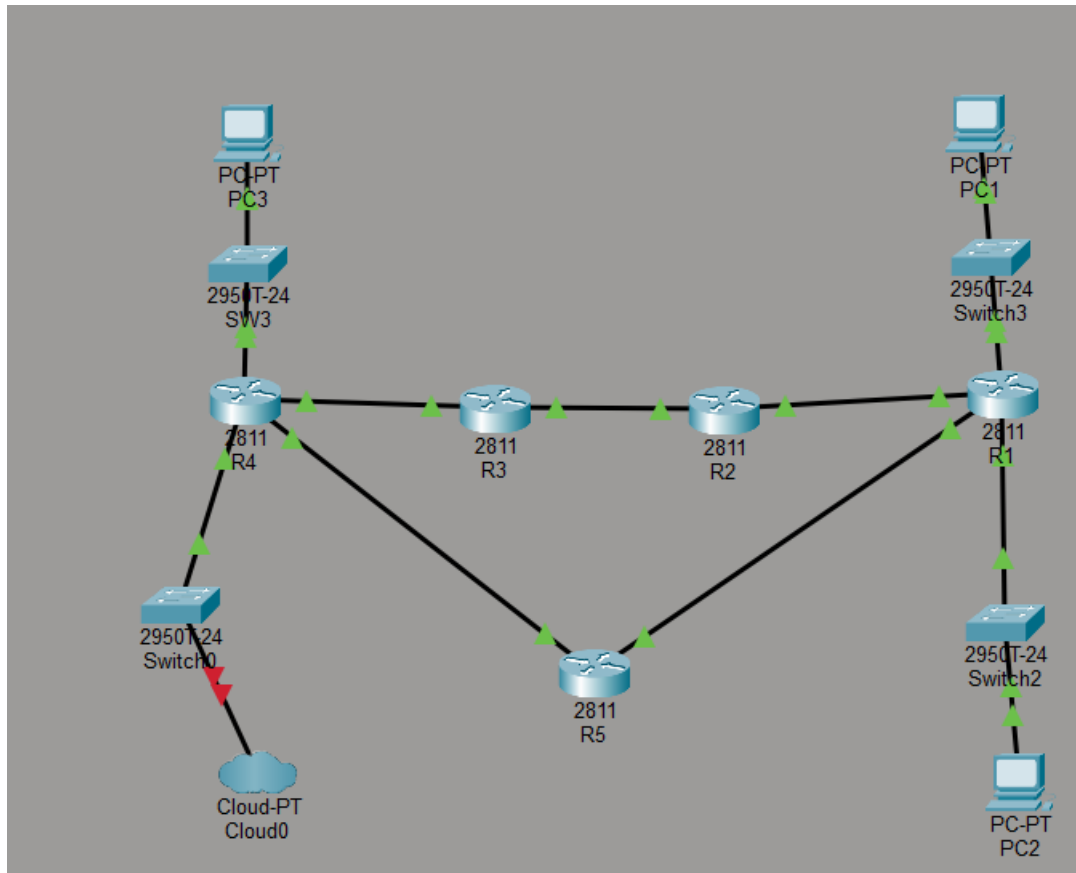


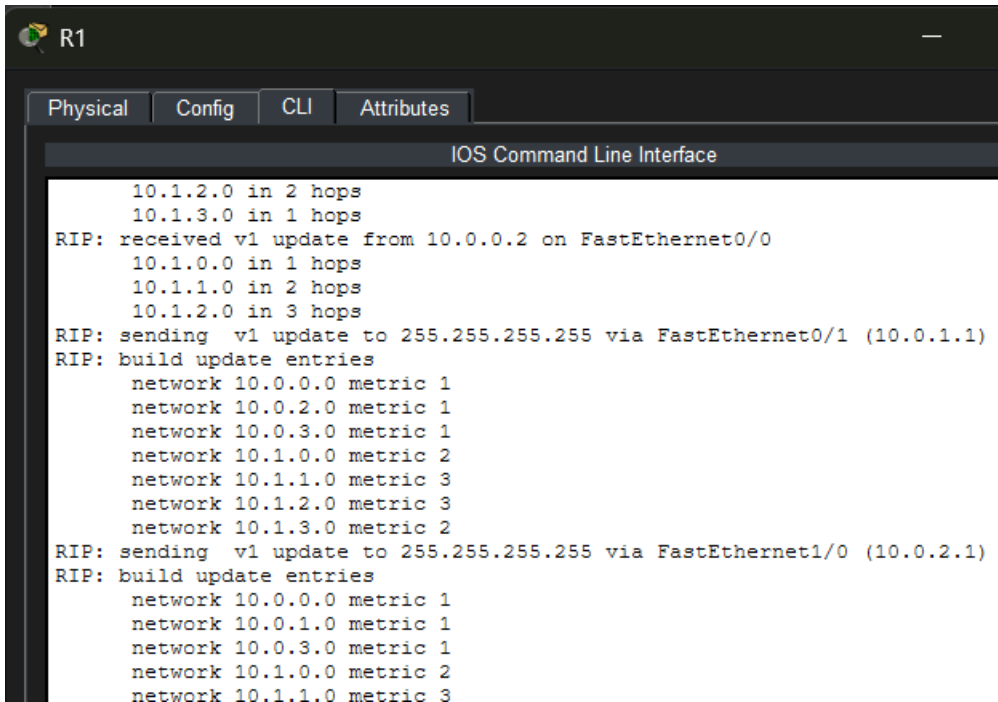
Topology



Task 74

```
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R2(config)#router rip
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R3(config)#router rip
R3(config-router)#network 10.0.0.0
R3(config-router)#no auto-summary
R4(config)#router rip
R4(config-router)#network 10.0.0.0
R4(config-router)#no auto-summary
R5(config)#router rip
R5(config-router)#network 10.0.0.0
R5(config-router)#no auto-summary
```

Task 75



```
R1
Physical Config CLI Attributes
IOS Command Line Interface

10.1.2.0 in 2 hops
10.1.3.0 in 1 hops
RIP: received v1 update from 10.0.0.2 on FastEthernet0/0
10.1.0.0 in 1 hops
10.1.1.0 in 2 hops
10.1.2.0 in 3 hops
RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (10.0.1.1)
RIP: build update entries
network 10.0.0.0 metric 1
network 10.0.2.0 metric 1
network 10.0.3.0 metric 1
network 10.1.0.0 metric 2
network 10.1.1.0 metric 3
network 10.1.2.0 metric 3
network 10.1.3.0 metric 2
RIP: sending v1 update to 255.255.255.255 via FastEthernet1/0 (10.0.2.1)
RIP: build update entries
network 10.0.0.0 metric 1
network 10.0.1.0 metric 1
network 10.0.3.0 metric 1
network 10.1.0.0 metric 2
network 10.1.1.0 metric 3
```

It sends a lot of updates to each interface available, the updates are broadcast.

Task 76

```
RIP protocol debugging is off
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#

R2>ena
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#

R3>ena
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#

R4>ena
R4#conf t
^
% Invalid input detected at '^' marker.

R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#

R5>ena
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#
```

Task 77

```

10.1.3.0/24 via 0.0.0.0 in 1 hops
RIP: sending v2 update to 224.0.0.9 via FastEthernet0/1 (10.0.1.1)
RIP: build update entries
    10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.2.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
    10.1.0.0/24 via 0.0.0.0, metric 2, tag 0
    10.1.1.0/24 via 0.0.0.0, metric 3, tag 0
    10.1.2.0/24 via 0.0.0.0, metric 3, tag 0
    10.1.3.0/24 via 0.0.0.0, metric 2, tag 0
RIP: sending v2 update to 224.0.0.9 via FastEthernet1/0 (10.0.2.1)
RIP: build update entries
    10.0.0.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.1.0/24 via 0.0.0.0, metric 1, tag 0
    10.0.3.0/24 via 0.0.0.0, metric 1, tag 0
    10.1.0.0/24 via 0.0.0.0, metric 2, tag 0
    10.1.1.0/24 via 0.0.0.0, metric 3, tag 0
    10.1.2.0/24 via 0.0.0.0, metric 3, tag 0
    10.1.3.0/24 via 0.0.0.0, metric 2, tag 0

```

Now it uses multicast instead of broadcast.

Task 78

```

RIP protocol debugging is off
R1#show ip rip database
10.0.0.0/24    auto-summary
10.0.0.0/24    directly connected, FastEthernet0/0
10.0.1.0/24    auto-summary
10.0.1.0/24    directly connected, FastEthernet0/1
10.0.2.0/24    auto-summary
10.0.2.0/24    directly connected, FastEthernet1/0
10.0.3.0/24    auto-summary
10.0.3.0/24    directly connected, FastEthernet1/1
10.1.0.0/24    auto-summary
10.1.0.0/24    [1] via 10.0.0.2, 00:00:03, FastEthernet0/0
10.1.1.0/24    auto-summary
10.1.1.0/24    [2] via 10.0.0.2, 00:00:03, FastEthernet0/0    [2] via 10.0.3.2, 00:00:03, FastEthernet1/1
10.1.2.0/24    auto-summary
10.1.2.0/24    [2] via 10.0.3.2, 00:00:24, FastEthernet1/1
10.1.3.0/24    auto-summary
10.1.3.0/24    [1] via 10.0.3.2, 00:00:24, FastEthernet1/1
R1#

```

Task 79

```

RIP protocol debugging is off
R1#show ip rip database
10.0.0.0/24    auto-summary
10.0.0.0/24    directly connected, FastEthernet0/0
10.0.1.0/24    auto-summary
10.0.1.0/24    directly connected, FastEthernet0/1
10.0.2.0/24    auto-summary
10.0.2.0/24    directly connected, FastEthernet1/0
10.0.3.0/24    auto-summary
10.0.3.0/24    directly connected, FastEthernet1/1
10.1.0.0/24    auto-summary
10.1.0.0/24    [1] via 10.0.0.2, 00:00:03, FastEthernet0/0
10.1.1.0/24    auto-summary
10.1.1.0/24    [2] via 10.0.0.2, 00:00:03, FastEthernet0/0    [2] via 10.0.3.2, 00:00:03, FastEthernet1/1
10.1.2.0/24    auto-summary
10.1.2.0/24    [2] via 10.0.3.2, 00:00:24, FastEthernet1/1
10.1.3.0/24    auto-summary
10.1.3.0/24    [1] via 10.0.3.2, 00:00:24, FastEthernet1/1
R1#

```

Task 80. Enabling OSPF on every router

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 10.0.0.0 255.255.255.0 area 0
^
% Invalid input detected at '^' marker.

R1(config-router)#network ?
  A.B.C.D Network number
R1(config-router)#network 10.0.0.0 255.255.255.0
% Incomplete command.
R1(config-router)#network 10.0.0.0 255.255.255.0 area 0
R1(config-router)#

R2>ena
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#network 10.0.0.0 255.255.255.0 area 0

R3>ena
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.0.0.0 255.255.255.0 area 0

R4>ena
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router ospf 1
R4(config-router)#network 10.0.0.0 255.255.255.0 area 0

R5>ena
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router ospf 1
R5(config-router)#network 10.0.0.0 255.255.255.0 area 0

```

Task 81.

```

R1#
R1#show ip route rip
    10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
R    10.1.0.0/24 [120/1] via 10.0.0.2, 00:00:07, FastEthernet0/0
R    10.1.1.0/24 [120/2] via 10.0.0.2, 00:00:07, FastEthernet0/0
    [120/2] via 10.0.3.2, 00:00:16, FastEthernet1/1
R    10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:16, FastEthernet1/1

```

R1#

Task 82

I predict the routing table will lose one route

```

R1>ena
R1#show ip route rip
    10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
R    10.1.0.0/24 [120/3] via 10.0.3.2, 00:00:03, FastEthernet1/1
R    10.1.1.0/24 [120/2] via 10.0.3.2, 00:00:03, FastEthernet1/1
R    10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:03, FastEthernet1/1

```

(I was actually right about it, it lost the FastEthernet0/0 routes.

Task 83

```

R1#show ip ospf database
        OSPF Router with ID (10.0.3.1) (Process ID 1)

        Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum Link count
10.1.0.2     10.1.0.2     200         0x80000003  0x00b170  1
10.0.3.1     10.0.3.1     66          0x80000004  0x0074d1  0
R1#

```

Task 84

Broadcast is used in RIPv1, which is changed to multicast in RIPv2. Which is faster and is less network traffic

Task 85

Static routes are manually configured, OSPF dynamically manages

Task 86

Yes, some are still available even though they may not be used. OSPF learns dynamically and adjusts the table, so it sees that there is one for itself via the static routes in place and doesn't need to create one, as it is by default trying to get the shortest path.

Task 87.

The routing table lost the FastEthernet0/0 routes, as in they were completely removed.

Task 88. RIP database has less routes in its database than the OSPF database, that is because RIP collects neighbours routes based on hop counts, where OSPF uses a different algorithm to find its shortest routes based on path cost. OSPF also has areas and can break down large topologies with these areas. OSPF builds a full map, RIP fetches neighbour routes.